Experiment–3: I-V Characteristics of Diode

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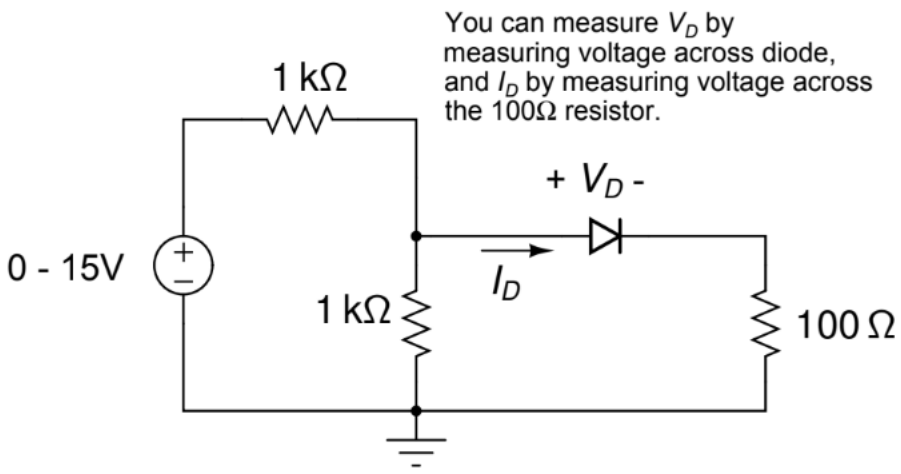
Part-1: Forward Characteristics of Various Diodes

Aim of the Experiment: - Observe and measure the forward bias I-V characteristics of a p-n Silicon diode (1N4007), Zener diode and LED.

Equipments required: -

- Required resistor values– 1 kΩ (×2), 100Ω (×1)
- Variable DC power supply (0-15V)
- 1N4007 Silicon diode, Zener diode, LED
- Multimeter
- Breadboard (×1)
- Connecting wires

Circuit Diagram: -

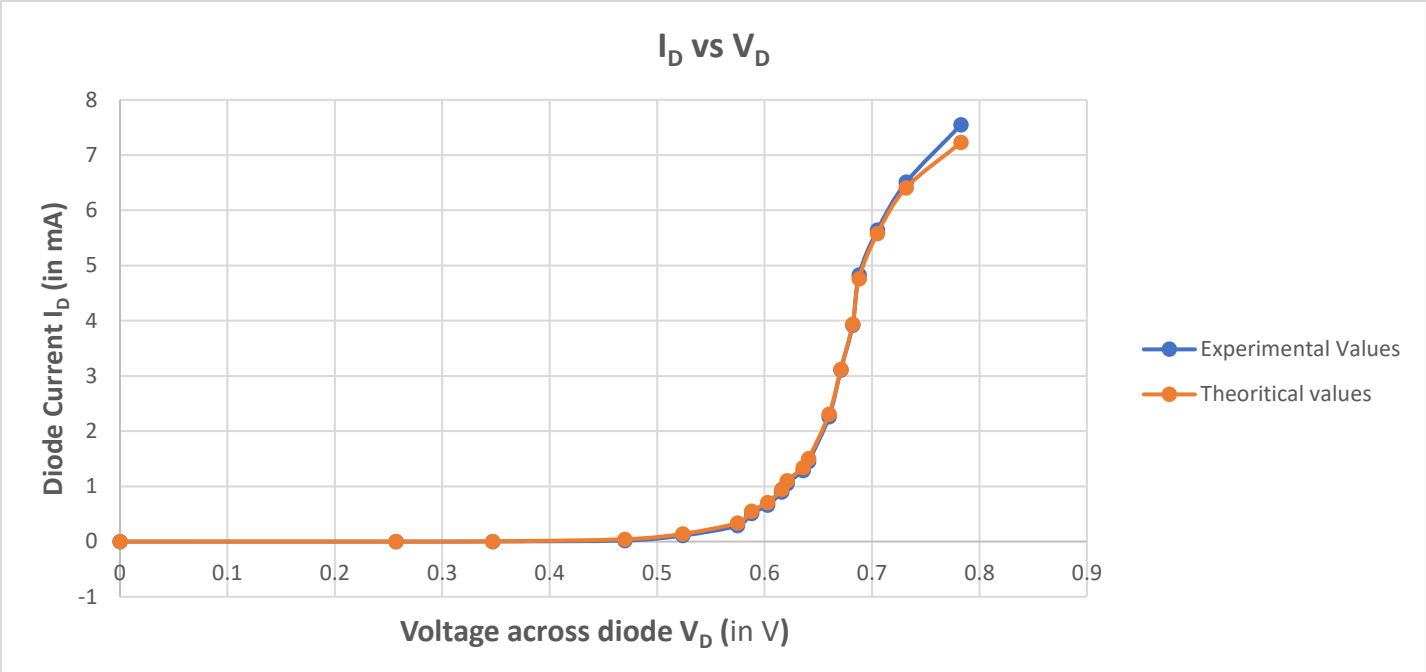


Part-1A: Forward Characteristics of Si Diode

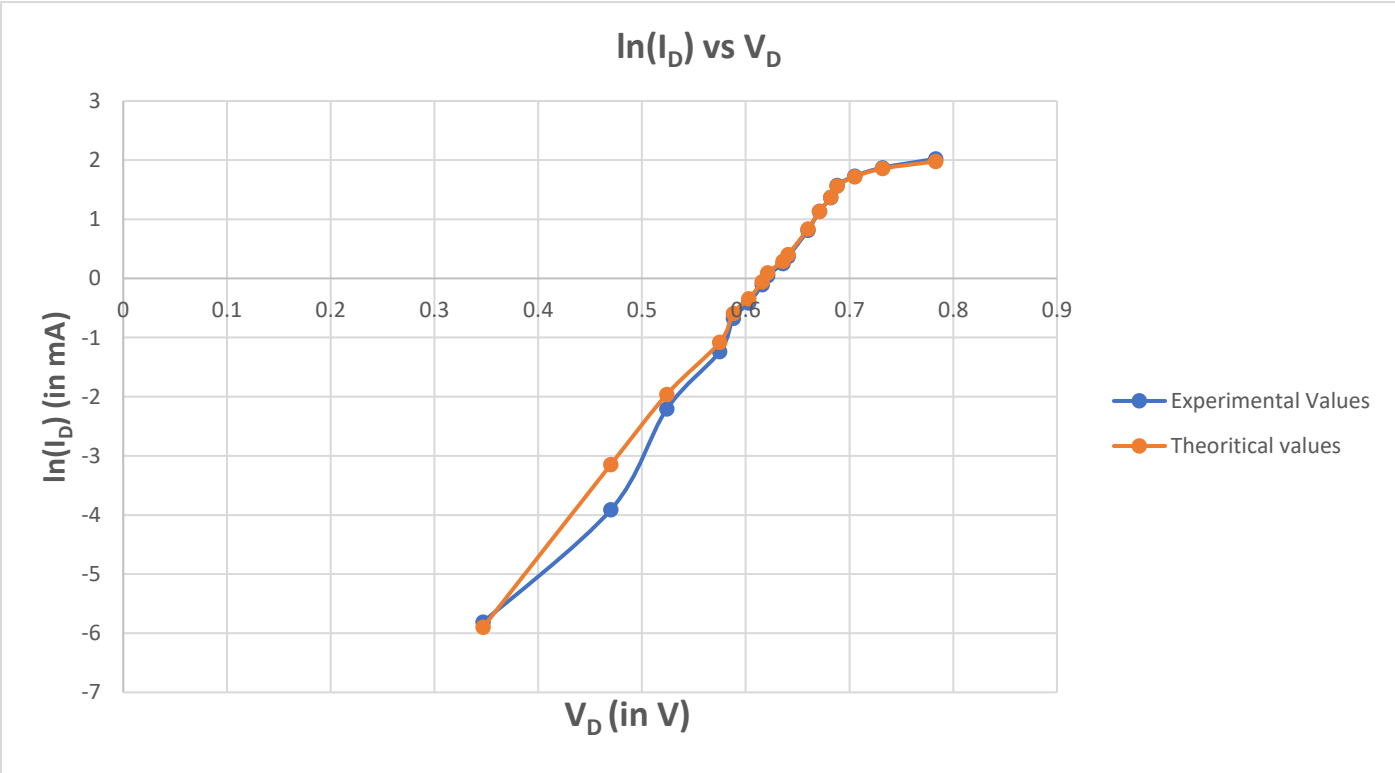
Observation Table: -

DC Supply (Volts)	Voltage across diode V_D (mV)	Voltage across 100Ω resistor V_R (mV)	Diode Current $I_D = V_R/100$ (mA)
0	0	0	0
0.5	257	0	0
0.7	347	0.3	0.003
1	470	2	0.02
1.2	524	11	0.11
1.5	575	29	0.29
1.8	588	51	0.51
2	603	66	0.66
2.3	616	90	0.9
2.5	621	105	1.05
2.8	636	129	1.29
3	641	145	1.45
4	660	226	2.26
5	671	311	3.11
6	682	392	3.92
7	688	483	4.83
8	705	564	5.64
9	732	585	6.51
10	783	625	7.55

Theoretically calculated vs observed graph values:-



Cut-in Voltage: 636 mV



Calculation: -

The voltage across the 100Ω resistor was measured using a multimeter and then the current through the diode was calculated using Ohm's Law:

$$I_D = \frac{VR}{R}$$

Forward resistance:

$$\boxed{r_{on} = \frac{dVd}{dId} \approx \frac{1}{m}}$$

Taking the two points **(0.641,1.5)** and **(0.682,3.94)** in the **ON** region of the curve we have the slope of the curve **m = 0.0595**

Hence, $r_{on} = 16.80\,\Omega$

From Shockley's equation:

$$I_D \approx I_S\, e^{\frac{Vd}{nVt}}$$

$$\therefore \ln(I_D) = \frac{Vd}{nVt} + \ln(I_S)$$

I_S = Reverse bias saturation current

n = Diode ideality factor V_T = Thermal Voltage ≈ 26mV

Taking the two points **(0.636,0.293)** and **(0.682,1.371)** from the graph of **ln(I_D) vs V_D** we have the slope of the graph at **ON** position equal to 0.0234 $(mV)^{-1}$.

Now, slope of the graph:

$$m = \frac{1}{n.Vt}$$

$$\text{or, } n = \frac{1}{m.Vt}$$

Hence, we get **n = 1.644.**

Taking the point **(0.66,0.837)** on the graph of **ln(I_D) vs V_D** and putting values on Shockley's equation. We get,

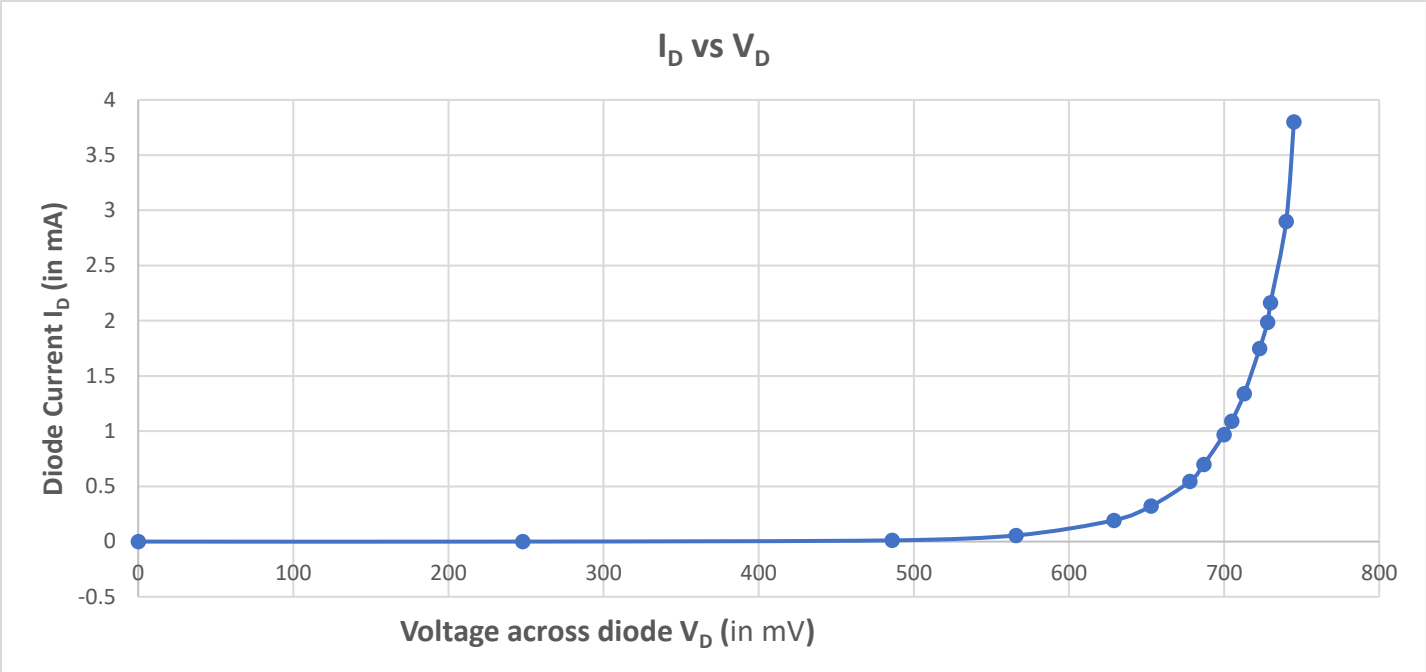
$I_S \approx$ 0.45 nA

Part-1B: Forward Characteristics of Zener Diode

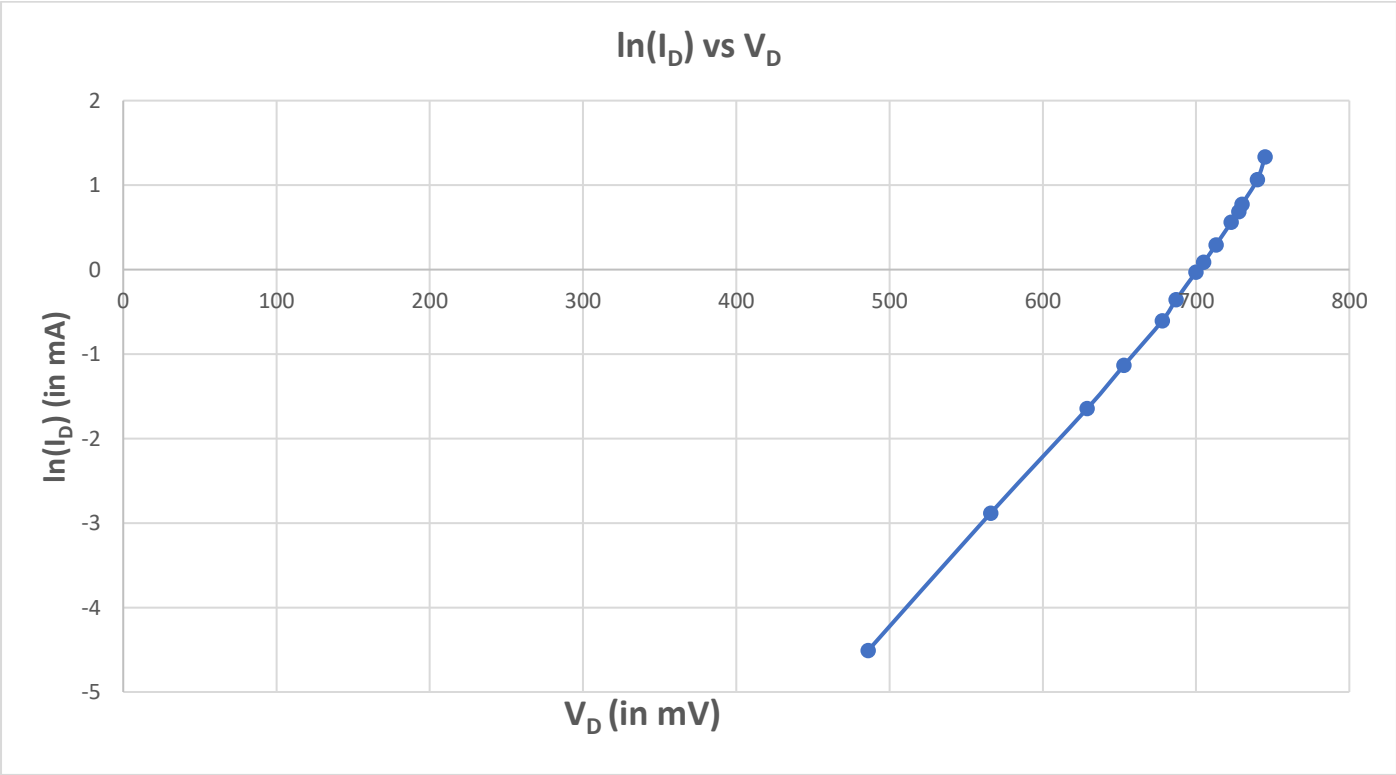
Observation Table: -

DC Supply (Volts)	Voltage across diode V_D (mV)	Voltage across 100Ω resistor V_R (mV)	Diode Current $I_D = V_R/100$ (mA)
0	0	0	0
0.5	248	0	0
1	486	1.1	0.011
1.2	566	5.6	0.056
1.5	629	19.3	0.193
1.7	653	32.2	0.322
2	678	54.6	0.546
2.2	687	70	0.7
2.5	700	97	0.97
2.8	705	109.2	1.09
3	713	134	1.34
3.5	723	175	1.75
3.8	728	198.5	1.985
4	730	216.4	2.164
5	740	290	2.9
6	745	380	3.8

Graphs:-



Cut-in Voltage: 687 mV



Calculation: -

The voltage across the 100Ω resistor was measured using a multimeter and then the current through the diode was calculated using Ohm's Law:

$$I_D = \frac{VR}{R}$$

Forward resistance:

$$\boxed{r_{on} = \frac{dVd}{dId} \approx \frac{1}{m}}$$

Taking the two points **(705,1.092)** and **(728,1.985)** in the **ON** region of the curve we have the slope of the curve **m = 0.0388**

Hence, $r_{on} = 25.76\,\Omega$

From Shockley's equation:

$$I_D \approx I_S\, e^{\frac{Vd}{nVt}}$$

$$\therefore \ln(I_D) = \frac{Vd}{nVt} + \ln(I_S)$$

I_S = Reverse bias saturation current

n = Diode ideality factor V_T = Thermal Voltage ≈ 26mV

Taking the two points **(566,-2.882)** and **(678,-0.605)** from the graph of **ln(I_D) vs V_D** we have the slope of the graph at **ON** position equal to 0.0203 $(mV)^{-1}$.

Now, slope of the graph:

$$m = \frac{1}{n.Vt}$$

$$\text{or, } n = \frac{1}{m.Vt}$$

Hence, we get **n = 1.892.**

Taking the point **(629,-1.645)** on the graph of **ln(I_D) vs V_D** and putting values on Shockley's equation. We get,

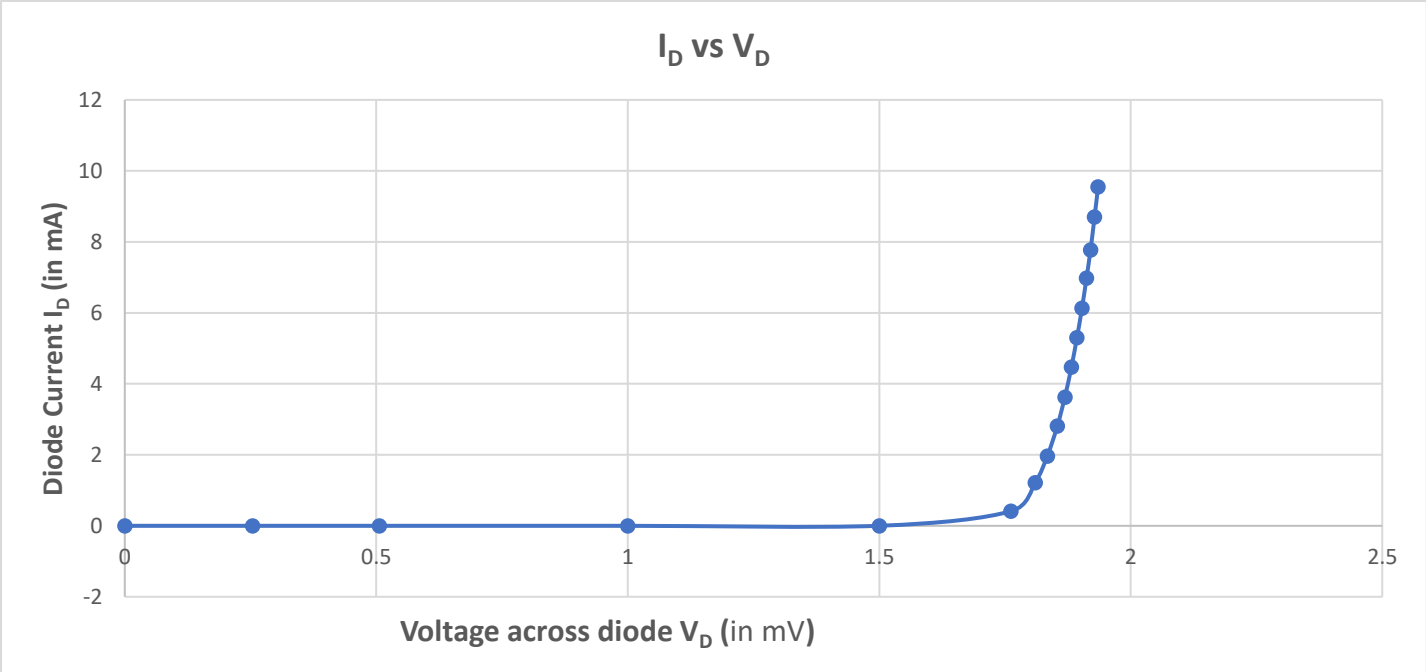
$I_S \approx 0.55$ nA

Part-1C: Forward Characteristics of LED

Observation Table: -

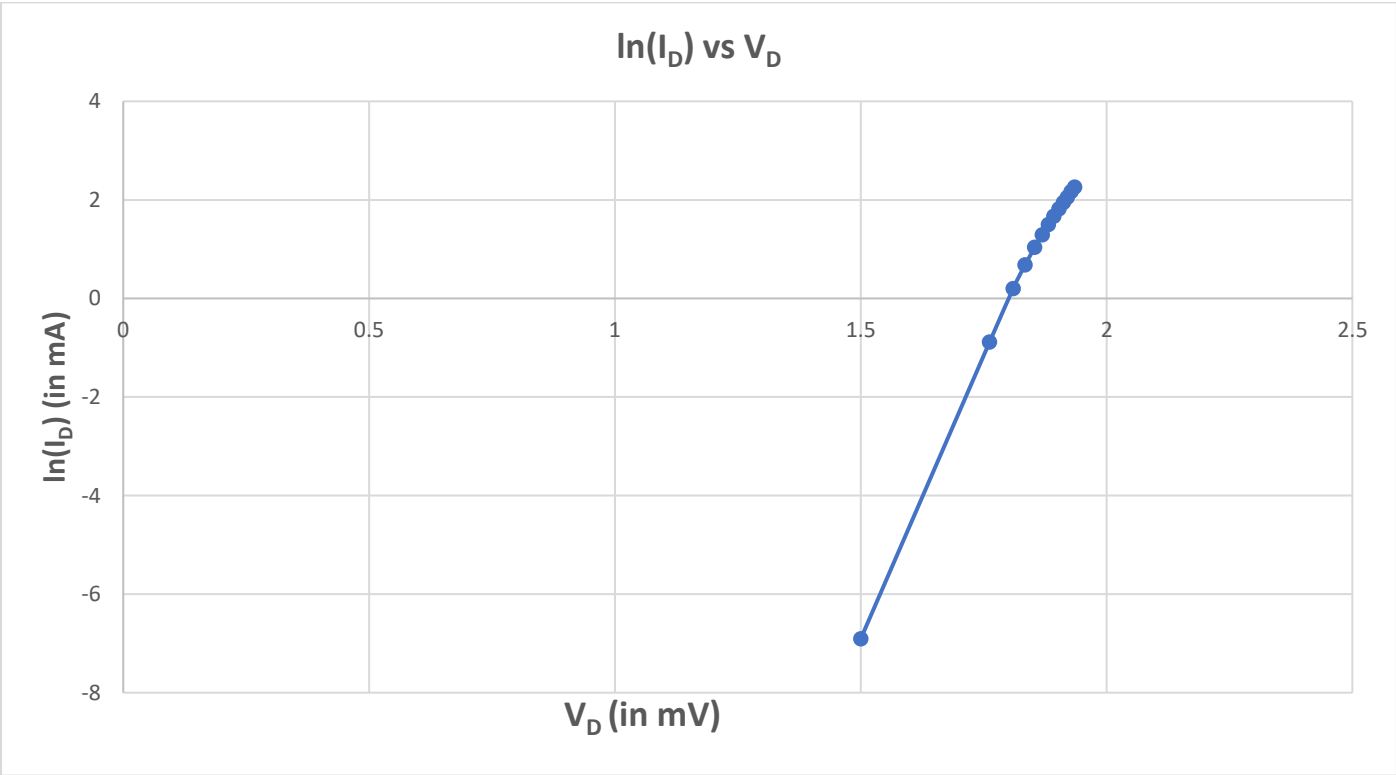
DC Supply (Volts)	Voltage across diode V_D (in V)	Voltage across 100Ω resistor V_R (mV)	Diode Current $I_D = V_R/100$ (mA)
0	0	0	0
0.5	0.254	0	0
1	0.506	0	0
2	1	0	0
3	1.5	0.1	0.001
4	1.762	41.2	0.412
5	1.81	122.1	1.221
6	1.834	196.4	1.964
7	1.854	280.9	2.809
8	1.869	362.9	3.629
9	1.882	447	4.47
10	1.893	530	5.3
11	1.903	613	6.13
12	1.912	698	6.98
13	1.92	777	7.77
14	1.928	870	8.7
15	1.935	955	9.55

Graphs:-



Cut-in Voltage: 1.762 V

Color of LED: Red



Calculation: -

The voltage across the 100 Ω resistor was measured using a multimeter and then the current through the diode was calculated using Ohm's Law:

$$I_D = \frac{VR}{R}$$

Forward resistance:

$$\boxed{r_{on} = \frac{dVd}{dId} \approx \frac{1}{m}}$$

Taking the two points **(1.854,2.809)** and **(1.92,7.77)** in the **ON** region of the curve we have the slope of the curve **m = 0.0752**

Hence, $r_{on} = 13.31\,\Omega$

From Shockley's equation:

$$I_D \approx I_S\, e^{\frac{Vd}{nVt}}$$

$$\therefore \ln(I_D) = \frac{Vd}{nVt} + \ln(I_S)$$

I_S = Reverse bias saturation current

n = Diode ideality factor V_T = Thermal Voltage ≈ 26mV

Taking the two points **(1.834,0.675)** and **(1.912,1.943)** from the graph of **ln(I_D) vs V_D** we have the slope of the graph at **ON** position equal to 0.0223 $(mV)^{-1}$.

Now, slope of the graph:

$$m = \frac{1}{n.Vt}$$

$$\text{or, } n = \frac{1}{m.Vt}$$

Hence, we get **n = 1.725.**

Taking the point **(1.882,1.5)** on the graph of **ln(I_D) vs V_D** and putting values on Shockley's equation. We get,

$I_S \approx$ 0.65 nA

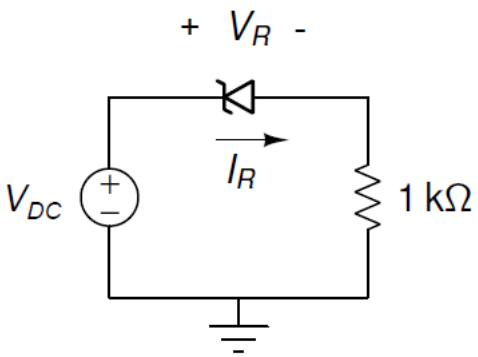
Part-2: Reverse Characteristics of Zener Diode

Aim of the Experiment: - Observe and measure the reverse bias I-V characteristics of Zener diode.

Equipments required: -

- Required resistor values– 1 kΩ (×1)
- Variable DC power supply (0-15V)
- Zener diode
- Multimeter
- Breadboard (×1)
- Connecting wires

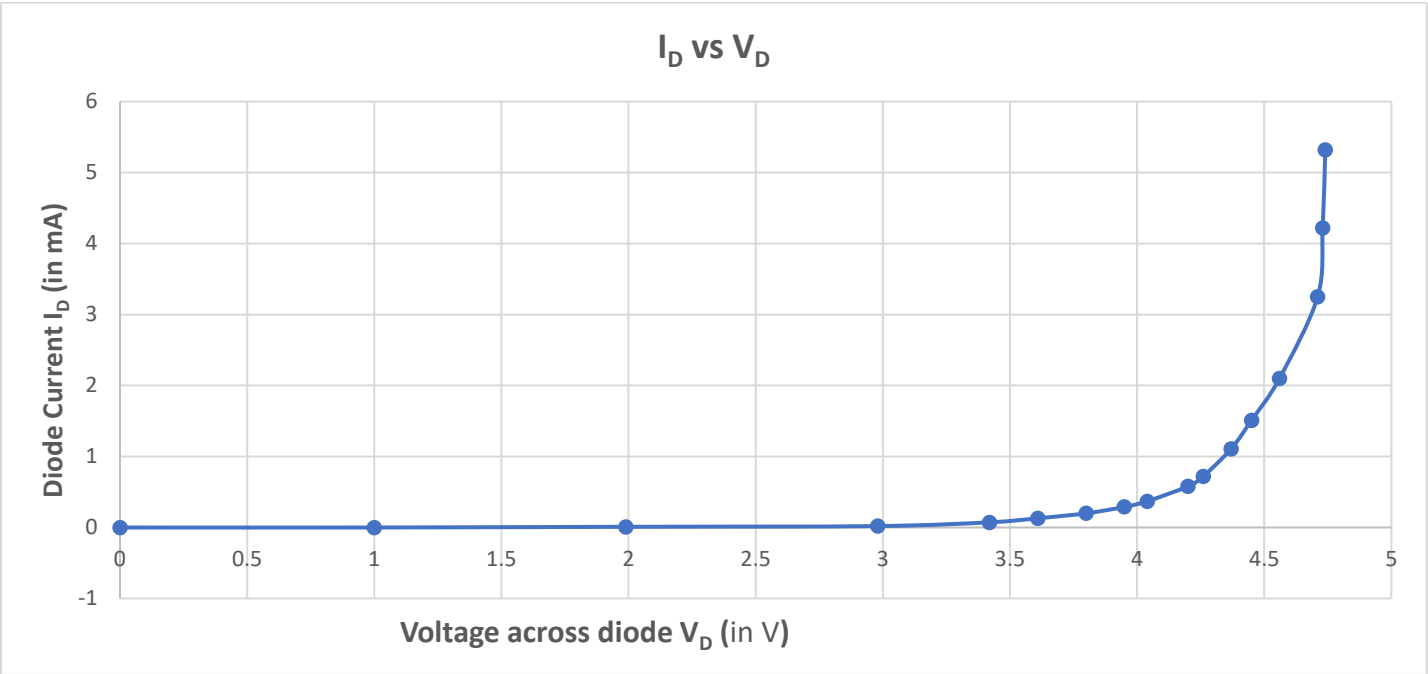
Circuit Diagram: -



Observation Table: -

DC Supply (Volts)	Diode reverse voltage V_D (in V)	Voltage across 1kΩ resistor V_R (in V)	Diode Current $I_D = V_R/1000$ (mA)
0	0	0	0
1	1	0	0
2	1.99	0.01	0.01
3	2.98	0.02	0.02
3.5	3.42	0.073	0.073
3.75	3.61	0.13	0.13
4	3.8	0.2	0.2
4.25	3.95	0.29	0.29
4.5	4.04	0.37	0.37
4.8	4.2	0.58	0.58
5	4.26	0.72	0.72
5.5	4.37	1.11	1.11
6	4.45	1.51	1.51
7	4.56	2.1	2.1
8	4.71	3.25	3.25
9	4.73	4.22	4.22
10	4.74	5.32	5.32

Graph:-



Reverse Breakdown Voltage: 4.26 V

Conclusion: -

- In this experiment we have practical understanding on how to use Si, Zener and LED Diode in forward and reverse bias in Zener diode.
- The input-output characteristics of diodes in forward and reverse bias were plotted and inference was drawn from those graphs on cut-in voltage and reverse breakdown voltage.
- Slight errors in the measurement of Voltage was observed, this was due to prolonged use of breadboard and the same 1k resistor. This resulting in heating up of circuit and thus the thermal tolerances of the circuit added to the error.

Discussion:-

➢ What is Shockley's equation?

The ***Shockley diode equation*** or the *diode law*, named after transistor co-inventor William Shockley of bell labs, models the exponential current-voltage I-V relationship of semiconductor diodes in moderate constant current forward bias or reverse bias.

It is given by the formula:

$$\boxed{\begin{aligned} &I_D \approx I_S\, e^{\frac{Vd}{nVt}} \\ &\therefore \ln(I_D) = \frac{Vd}{nVt} + \ln(I_S) \end{aligned}}$$

I_D = Diode current

V_D = Voltage across diode

I_S = Reverse bias saturation current

n = Diode ideality factor

V_T = Thermal Voltage $\approx$ 26mV

Shockley's equation is not valid for reverse-biased Zener diodes.

- **What does ideality factor represent in Shockley's equation?**

  The ideality factor (n) in the Shockley diode equation represents a correction factor that accounts for deveiation from ideality in a real diode's behavior. In other words, it quantifies how closely a real diode's behavior matches the ideal diode behavior described by the equation. The value of idelaity factor varies from 1 to 2. 1 representing an ideal diode.